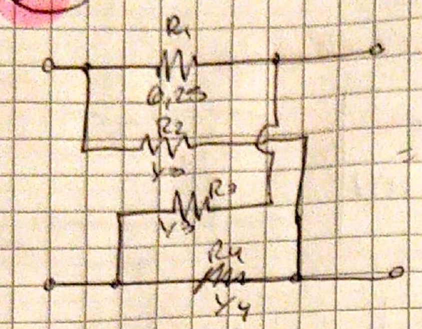
② Transformar la lattice en T puenteado.



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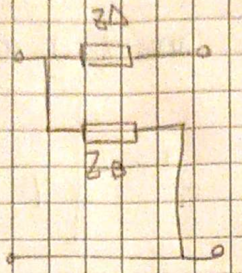
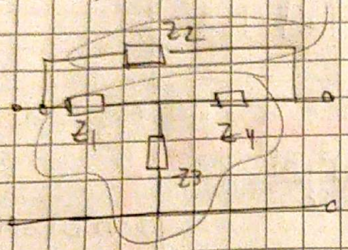


$$\Rightarrow Z = \begin{bmatrix} \frac{1}{Z_4} & \frac{1}{Z_4} \\ \frac{1}{Z_4} & \frac{1}{Z_4} \end{bmatrix}$$

$Z_A = Z_D \wedge Z_B = Z_C$



Como podemos tener la matriz Z del T sin puentear más fácil y trabajarla en admitancia para sumarla con la Z2 ya que estarían en paralelo

$$Z_{normal} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_3 + Z_4 \end{bmatrix} \Rightarrow Y_1 = \begin{bmatrix} \frac{Z_3+Z_4}{\Delta Z} & -\frac{Z_3}{\Delta Z} \\ -\frac{Z_3}{\Delta Z} & \frac{Z_1+Z_3}{\Delta Z} \end{bmatrix}$$

$$\Delta Z = (Z_1+Z_3)(Z_3+Z_4) - Z_3^2$$
$$Z_1Z_3 + Z_1Z_4 + \cancel{Z_3^2} + Z_3Z_4 - \cancel{Z_3^2}$$

$$T_{Z2} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \Rightarrow Y_{Z2} = \begin{bmatrix} Y_2 & -Y_2 \\ -Y_2 & Y_2 \end{bmatrix}$$

$$Y_{TOTAL} = [Y_1] + [Y_2] = \begin{bmatrix} \frac{Z_3+Z_4}{\Delta Z} + Y_2 & -\frac{Z_3}{\Delta Z} - Y_2 \\ -\frac{Z_3}{\Delta Z} - Y_2 & \frac{Z_1+Z_3}{\Delta Z} + Y_2 \end{bmatrix} = \begin{bmatrix} \frac{Z_3+Z_4}{Z_1Z_3+Z_1Z_4+Z_3Z_4} + \frac{1}{Z_2} & \frac{-Z_3}{Z_1Z_3+Z_1Z_4+Z_3Z_4} - \frac{1}{Z_2} \\ \frac{-Z_3}{Z_1(Z_3+Z_4)+Z_3Z_4} - \frac{1}{Z_2} & \frac{Z_1+Z_3}{Z_1Z_3+Z_1Z_4+Z_3Z_4} + \frac{1}{Z_2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{(Z_3+Z_4)Z_2 + Z_1(Z_3+Z_4) + Z_3Z_4}{[Z_1(Z_3+Z_4) + Z_3Z_4]Z_2} & -\left(\frac{Z_3Z_2 + Z_1(Z_3+Z_4) + Z_3Z_4}{Z_2(Z_1Z_3+Z_1Z_4+Z_3Z_4)}\right) \\ -\left(\frac{Z_3Z_2 + Z_1(Z_3+Z_4) + Z_3Z_4}{Z_2(Z_1Z_3+Z_1Z_4+Z_3Z_4)}\right) & \frac{(Z_1+Z_3)Z_2 + Z_1(Z_3+Z_4) + Z_3Z_4}{Z_2(Z_1Z_3+Z_1Z_4+Z_3Z_4)} \end{bmatrix}$$

$\rightarrow Z_{11} = \frac{y_{22}}{\Delta y}$ $\Rightarrow Z_{12} = \frac{-y_{12}}{\Delta Y}$ $\rightarrow Z_{21} = \frac{-y_{21}}{\Delta Y}$ $\rightarrow Z_{22} = \frac{y_{11}}{\Delta Y}$